

Arthrospira (Spirulina) platensis supplementation affects folliculogenesis, progesterone and ghrelin levels in fattening pre-pubertal gilts

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Received: 5 April 2017 / Revised and accepted: 29 August 2017 / Published online: 12 September 2017
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Abstract The aim of the present investigation was to study the effect of *Arthrospira (Spirulina) platensis* supplemented diet on follicular development and related endocrine parameters, such as estradiol and progesterone levels as well as ghrelin levels in pre-pubertal gilts. Twenty-one 60-day-old Danube White gilts, randomly divided in three groups—control and two experimental—were involved in the experiment. The diet of experimental animals was supplemented with 2 and 3 g of dry biomass/capita daily of the micro alga *A. platensis* (SP), respectively, during 120 days. Pigs weighing between 109.3 and 114.7 kg were slaughtered at the age of 180 days. The visual macroscopic and histological analysis of removed ovaries was done. Progesterone and estradiol levels by radioimmunoassay (RIA) and ghrelin by ELISE methods were measured. The presence of corpora lutea in the ovaries of animals from all three groups as well as the progesterone and estradiol levels indicated that the animals were cycling and puberty had started at the time of slaughtering. In the SP 3-g group, the number of large follicles and corpora lutea was significantly higher compared to the control group and the corpora albicans were found. A positive correlation was observed between ghrelin and progesterone among all groups as well as between

estradiol and ghrelin in the experimental groups independent on dosages of SP. *Arthrospira platensis*-supplemented diet provoked a successful follicular development and an earlier ovulation during sexual maturation of pigs between 60 and 180 days.

Keywords *Arthrospira platensis* · Pre-pubertal pigs · Folliculogenesis · Ghrelin · Progesterone · Estradiol

Introduction

Phytogenic feed additives such as a wide variety of herbs, spices, seaweed, and products derived thereof are more and more used as non-antibiotic growth promoters in livestock animals (Windisch et al. 2008; Thacker 2013). Several studies have shown that in pigs dietary supplementation with biologically active additives ensures an increase in production efficiency by improving the growth rate, feed utilization, and meat quality (Namkung et al. 2004; Oetting et al. 2006). These feed additives influence for instance gut microbiota composition and have bacterial effects on the gut (Yan et al. 2012; Li et al. 2012; Ahmed et al. 2013). They improve appetite leading to an increased feed intake (Costa et al. 2007), stimulating digestive secretions (mucus), and increase digestive enzymatic activity (Platel and Srinivasan 2004). The microalgae *Arthrospira platensis* (commonly called *Spirulina platensis*) is an example of such a phytogenic feed additive, which contains a combination of minerals, vitamins, essential amino acids, fatty acids, and antioxidants. The effects of dietary supplementation of *A. platensis* on animal productivity and health have been investigated in more detail in chicken, pigs, ruminants, and rabbits (Peiretti and Meineri 2011; Mariey et al. 2012; Holman and Malau-Aduli 2013).

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Arthrospira platensis further enhances the growth rate in weanling and fattening pigs and improves the microbial gut flora and meat quality (Hugh et al. 1985; Simkus et al. 2008; Saeid et al. 2013; Šimkus et al. 2013). It is therefore not surprising that more than half of the *A. platensis* produced worldwide is used in livestock and fish feeds (Belay et al. 1996; Belay 2002; Abdulrahman and Hamad 2014). Limited data are available on effects of dietary supplementation of *A. platensis* on reproductive parameters. A positive effect on reproductive performance has been reported in zebrafish and birds (Nikodémusz et al. 2010; Geffroy and Simon 2013). In boars, *A. platensis* supplementation leads to an increase in ejaculate volume and concomitantly an increase in sperm concentration and improvement in sperm mobility (Kistanova et al. 2009a; Rudic et al. 2008). Not much is known about the effects of *A. platensis* on female reproduction in mammals, except that when sows are fed the microalgae during pregnancy the birth weight of the newborn piglets was increased by approximately 20% (Simkus et al. 2009). It may be clear from the above that reproductive function is strongly influenced by various nutritional and metabolic factors. The hormone ghrelin plays an important role in the control of food intake and energy balance and reproduction (Govoni et al. 2005). Ghrelin acts at different levels of the reproductive system: it influences pituitary hormone release but also acts directly on the gonads affecting tissue development and steroidogenesis (Tena-Sempere 2007; Tena-Sempere et al. 2013). These observations were confirmed in vitro using porcine granulosa cells (Sirotkin et al. 2011; Rak-Mardyla et al. 2012). Ghrelin is also expressed in the porcine ovary; messenger RNA (mRNA) levels appear to be dependent on the stage of the estrus cycle (Zhang et al. 2008). In line with these observations, Torres-Rovira et al. (2011) showed in the Iberian pig that the concentration of ghrelin is significantly lower in animals in estrus than in animals failing to display estrus behavior. Whether ghrelin also plays a role in puberty onset, especially in pig, and should be affected by nutritional factors, is at present not known.

The aim of the present investigation was to establish whether a prolonged period of fortification of the diet of prepubertal gilts with *A. platensis* has an effect on follicular development and related endocrine parameters, such as estradiol and progesterone levels as well as ghrelin levels in prepubertal gilts.

Material and methods

Animals

The animal experiment was performed at the Experimental station in Agricultural Institute Shumen, Bulgaria. All animals were reared and handled in accordance with the Bulgarian

veterinary law related to the life conditions and welfare of livestock experimental animals (Regulation N20/01.11.2012 based on the Directive 2010/63/EU on the protection of animals used for scientific purposes in the Member States). The use of animals in the experiment was approved by the Ethics Board in Agricultural Institute-Shumen. Pigs were raised at 8–20 °C temperature and 65% humidity. Delousing of animals was made by IVOMEK.

Twenty-one 60-day-old Danube White gilts weighing 34.7 to 34.8 kg were randomly divided over three groups of seven gilts, housed in three pig pens. The diet of the gilts was adapted in accordance with two phases of growth. Between 30 and 80 kg of body weight, the animals were fed a diet consisting of 17.24% crude protein, 13.30 MJ digestible energy, and 12.81 MJ metabolizable energy (Table 1). Between a body weight of 80 and 110 kg, the gilts were fed a diet 15.7% crude protein, 13.48 MJ digestible energy, and 12.93 MJ metabolizable energy. The gilts in group I received this diet without any further additions; to the diet of the gilts in groups II and III 2 and 3 g of 100% dry biomass (at 40 °C)/capita daily of the cyanobacterium *Arthrospira platensis* (SP), respectively (Bioagrovet product, Lithuania), by individual feeding. The content of bioactive compounds in *A. platensis*, used in the experiment, is presented in previous publications (Nedeva et al. 2014). The dietary intervention lasted 120 days; pigs weighing between 109.3 and 114.7 kg were slaughtered at the age of 180 days.

Slaughter procedure was carried out in the slaughterhouse according to Regulation No. 22 (14th of December, 2005—“reducing animal suffering during slaughtering and killing”) by the people entitled to professional slaughter. Approved procedure involves use of electric shock.

Chemical composition of feed samples was established by the acquired methods in Agricultural Institute-Shumen (Sandeve 1979). Protein was established by the Kjeldahl method, crude fats—by the use of Soxhlet extractor, fibers—by the Weende method. Calcium was established by the oxalate method, phosphorus—by the molybdate method of Gerike and Kurmis (Sandeve 1979). Digestive energy (DE) and metabolizable energy (ME) were calculated on the basis of digestive nutrients from our investigations and the use of the following equations:

$$\begin{aligned} \text{DE} = & 5.78 \times \text{Dig.Protein (g)} + 9.42 \times \text{Dig.Fats (g)} \\ & + 4.40 \times \text{Dig.Fibers (g)} + 4.07 \\ & \times \text{Dig.Non-nitrogenous extract substances (g)}. \end{aligned}$$

$$\begin{aligned} \text{ME} = & 5.01 \times \text{Dig.Protein (g)} + 8.93 \times \text{Dig.Fats (g)} \\ & + 3.44 \times \text{Dig.Fibers (g)} + 4.08 \\ & \times \text{Dig.Non-nitrogenous extract substances (g)}. \end{aligned}$$

Table 1 Nutrients content in 1 kg of compound feed

Ingredients	30–80 kg live weight	80–110 kg live weight
Water (%)	12.31	12.30
Crude protein (%)	17.24	15.17
Crude fiber (%)	5.91	6.78
Crude ash (%)	4.97	3.35
Crude fats (%)	2.63	2.72
Lysine (%)	0.83	0.71
Methionine (%)	0.32	0.30
Calcium (%)	1.01	0.61
Phosphorus (%)	0.68	0.56
Sodium (%)	0.12	0.13
Cu (mg/kg)	157.5	131.25
Vitamin A (I.U. kg ⁻¹)	10,500	8750
Vitamin E/Dh α -tocopherol (mg kg ⁻¹)	90	75
Vitamin D ₃ /cholecalciferol (I.U. kg ⁻¹)	2000	1750

Pigs were fed with compound feed and water ad libitum.

The evening before slaughter, the feed was taken away. In the morning of the day of slaughter, blood samples (3 mL) were collected by jugular puncture using heparinized tubes. Plasma was processed for the analysis of the following hormones: progesterone, estradiol-17 β , and ghrelin.

Histology

After slaughtering, the ovaries were removed and examined macroscopically. Follicles were divided in two classes, 2–6 and > 6 mm, counted, and measured. The number of corpora lutea was also counted. Next, the ovaries were fixed in 4% formalin for at least 48 h. The ovaries were cut into six to eight pieces and embedded in paraffin. Five-micrometer sections were cut from different pieces. Sections were stained with PAS and Mayer's hematoxylin and examined using an Olympus BX51 microscope (Tokyo, Japan).

Hormone analysis

Progesterone and estradiol levels were measured by the radioimmunoassay (RIA) method in the certificated laboratory for veterinary diagnostic. The assay was based on the competition between unlabelled progesterone/estradiol and a fixed quantity of ¹²⁵I-labeled progesterone/estradiol for a limited number of binding sites on progesterone/estradiol-specific antibodies (Cisbio Bioassays). The progesterone assay range was 0–37.7 ng mL⁻¹, and cross reactions with progesterone was 100%, with 17- β -estradiol < 0.001; the intra-assay coefficient of variation was 3.5–4.4%, and extra-assay 4.5–4.0%. The measuring range of estradiol was 5000 pg mL⁻¹, and cross reactions with 17- β -estradiol was 100%, with progesterone

< 0.004; the intra-assay coefficient of variation was 3.5–4.2%, and extra-assay 4.9–4.8%.

Ghrelin levels were analyzed using a commercial ELISA kit (E90991Po 96 Tests, Life Sciences Inc., PRC) following the manufacturer's instructions. The developed samples were analyzed using an ELISA microplate reader at wavelength 450 nm (LKB, Italy).

Statistics

For data analysis, Statistica Ver.10 was used. The results are presented as mean and standard error of mean (SEM). Parametric (Student *t* test) and non-parametric (Mann-Whitney *U* test) statistical methods were applied in dependence on sample number, normality of distribution, and equality of variance for the estimation of difference significance.

Results

In the present study, it was investigated whether SP can affect ovarian development in pre-pubertal gilts. In order to compare the different treatment groups, all gilts were slaughtered at the same age. When slaughtered, it appeared that the gilts were already cycling, inducing variation in progesterone and estradiol levels as not all gilts were slaughtered at the same stage of the estrus cycle.

The addition of *A. platensis* (2 and 3 g per capita daily) in the compound feed of fattening pigs leads to the tendency to increase the growth intensity by 2.13–4.57% and to a better feed conversion ratio by 3.6% (Table 2).

The body weight at the end of the experiment was not significantly different among the three groups, though body weight tended to be somewhat increased in the SP 3-g group

Table 2 Morphometric parameters of reproductive organs and fattening traits of animals

Parameters	Control	SP 2 g	SP 3 g
Body weight	110.0 ± 1.4	110.6 ± 0.7	114 ± 1.6
Average periodic feed intake:			
1st (30–80 kg)	2.6 ± 0.06	2.6 ± 0.04	2.6 ± 0.01
2nd (80–110 kg)	3.3 ± 0.03	3.3 ± 0.02	3.3 ± 0.03
Average feed intake	2.8 ± 0.05	2.8 ± 1.5	2.8 ± 0.6
Periodic feed conversion ratio:			
1st (30–80 kg)	4.0 ± 0.35	4.1 ± 0.39	3.8 ± 0.36
2nd (80–110 kg)	4.8 ± 0.31	4.4 ± 0.78	4.4 ± 0.45
Feed conversion ratio	4.2 ± 0.32	4.2 ± 0.43	4.0 ± 0.36
Average periodic daily weight gain:			
1st (30–80 kg)	645 ± 3.2	630 ± 2.9	687 ± 3.1
2nd (80–110 kg)	680 ± 2.5	762 ± 4.5	680 ± 2.4
Average daily weight gain	657 ± 2.9	671 ± 2.3	687 ± 2.9
Cervix length	10.7 ± 0.9	13.1 ± 0.7	13.7 ± 1.3
Length left uterine horn	85.2 ± 9.6	103.3 ± 8.8	87.3 ± 2.7
Length right uterine horn	80.9 ± 10.9	95.1 ± 8.6	81.6 ± 2.8
Lumen left uterine horn	3.1 ± 0.4	3.1 ± 0.4	3.7 ± 0.4
Lumen right uterine horn	2.9 ± 0.3	3.4 ± 0.4	3.2 ± 0.4

Numbers of animals are seven per group. Body weight, average feed intake per capita daily, and feed conversion ratio are expressed in kilograms, average daily weight gain is expressed in grams, and cervix, uterine horn, and lumen uterine horn are expressed in centimeters. No significant differences were observed among the three groups. Data are expressed as mean ± SEM; statistical analysis was performed by Mann-Whitney *U* test (*number N* = 7 is too small to form a normal distribution) *SP Arthrospira platensis* dosages

(Table 2). No significant differences in uterus horn length or lumen diameter were observed. The cervix length did also not differ among the groups (Table 2).

Macroscopical analysis of the ovaries showed that the number of follicles in the 2–6- and > 6-mm size classes did not differ between the control and the SP 2-g groups (Table 3). In contrast, in the SP 3-g group, the number of follicles in the 2–6-mm size class was significantly lower compared to the control group, while the number of follicles > 6 mm was significantly higher (Table 3). The corpora lutea were found in the ovaries of all three groups of animals, indicating that the animals were cycling and puberty had started. In the SP 3-g group, the number of corpora lutea was however significantly higher compared to the control group (Table 3).

Histological analysis confirms the presence of large antral follicles in the ovaries of the three experimental groups (Fig. 1a). In the ovaries of the control group, there were also luteinized unruptured follicles observed, in which the oocyte had resumed the first meiotic division. The granulosa and theca cells in these follicles had undergone luteinization but failed to ovulate (Fig. 1b).

Table 3 Macroscopical analysis of the ovaries of 180-day-old gilts at the time of slaughter

	Follicles 2–6 mm	Follicles > 6 mm	Corpora lutea
Control	21.3 ± 1.1	4.1 ± 0.6	3.2 ± 0.3
SP 2 g	22.2 ± 1.8	4.5 ± 0.5	4.1 ± 0.3
SP 3 g	10 ± 1.1*	8.5 ± 0.6*	7 ± 0.4 ^a

^a *P* < 0.05 compared with the control group (Student *t* test applied for the statistical analysis due to normal distribution and equal variance). Gilts received either no treatment (control) or were treated daily with 2 or 3 g SP for 120 days. Antral follicles were divided in two classes: 2–6 and > 6 mm. All macroscopically visible follicles in these size classes were counted as were the corpora lutea. Values are expressed as mean ± SEM per ovary

The presence of the corpora lutea is confirmed at the histological levels in all three experimental groups (Fig. 2a, c). In the ovaries of the SP 3-g group, the remains of the degenerated corpora lutea (corpora albicans) were observed as well (Fig. 2b, d), suggesting that these gilts had undergone at least two estrous cycles. The corpora albicans was not observed in the ovaries of the control and SP 2 g groups.

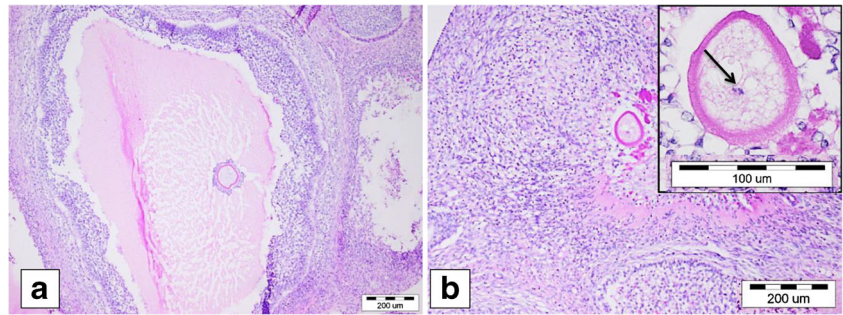
To confirm the presence of developing follicles and corpora lutea, plasma levels of estradiol and progesterone were determined. Estradiol levels tended to be lower in the SP 2-g and SP 3-g groups compared to the controls, but this did not reach the level of significance (Table 4). Plasma progesterone levels in the control gilts were low but measurable, confirming the presence of functional corpora lutea. Progesterone levels in the SP 2-g and SP 3-g groups were significantly higher compared to the controls. The difference in progesterone levels between the SP 2-g and SP 3-g groups was relatively small and not in line with the 75% higher number of the corpora lutea presents in the SP 3-g group (Table 4). The relatively large variation in progesterone and estradiol levels among the three groups of animals is due to the fact that the animals were slaughtered at different stages of the estrus cycle.

Plasma ghrelin levels were significantly increased in the SP 2-g group, while in the SP 3-g group a trend towards an increase was observed, compared to the age-matched control group (Table 4). A correlation assay was performed between ghrelin and estradiol levels; a correlation was observed between ghrelin and estradiol independent of the dose of SP (Fig. 3). A positive correlation was observed between ghrelin and progesterone among all groups: control *R* = 0.94, *P* < 0.01; SP 2 g *R* = 0.86, *P* < 0.01; SP 3 g *R* = 0.79, *P* < 0.05.

Discussion

The presented results had shown an earlier occurrence of the ovulation at the end of the experiment in the SP-treated groups. The data of hormone assay is in support of this.

Fig. 1 **a** Representative example of a large antral follicle in the ovary of a 180-day-old gilt treated daily for 120 days with 3 g SP. **b** Example of a luteinized unruptured follicle in the ovary of a 180-day-old control gilt. Inset shows a detail of the oocyte with mitotic figure (arrow). Bar represents 200 μm (**a**, **b**) and 100 μm (**b**, inset)



There was an increase of the progesterone level in both treated groups more than 1 nmol L^{-1} , as an evidence of puberty onset. The histological observation confirmed the presence of healthy follicles and corpora lutea in the ovary of experimental animals. Moreover, the corpora albicans in the 3-g SP group were found. The established high correlation between progesterone level and ghrelin in all groups supposes the active participation of this hormone in the sexual maturation of gilts. Additionally, the SP treatment leads to the establishment of the correlation between ghrelin and estradiol in both experimental groups.

There are many investigations related to the dietary use of SP in pigs for enhancement of productivity and improvement of healthy status, but no data about its effect on the reproductive system in gilts. Our results are the first evidence of the *Arthrospira* beneficial effect on folliculogenesis in pig ovary during sexual maturation. More active and successful folliculogenesis in SP-treated pigs was confirmed by hormonal assay, primary visual observation, and histological investigations of ovaries.

It is necessity underlining the beneficial effect of *Arthrospira* whole dry biomass use, because the chemical composition of this alga includes proteins (55–70%), carbohydrates (15–25%), essential fatty acids (18%), vitamins, minerals, and pigments with high antioxidative properties like carotenes, chlorophyll *a*, and phycocyanin (Sánchez et al. 2003; Nedeva et al. 2014). Not only energy balance of diet

is important for the valuable sexual maturation. The control of folliculogenesis, maturation, and developmental capacity of oocytes may be more complex than the provision of adequate energy for cellular processes. Many researches underline that the balanced intracellular redox status is crucial for successful development of oocytes during maturation (Dumollard et al. 2007; Agarwal et al. 2008; Shkolnik et al. 2011). Pre-pubertal pig oocytes are less competent in regulating redox balance compared to adult oocytes (Yuan et al. 2012). The modulating of the antioxidative status by addition of antioxidants should provoke the oocyte maturation. So a single injection of retinyl palmitate 6 days before mating in pigs increased the percentage of oocytes that progressed to metaphase II in vitro and promoted follicular homogeneity (Whaley et al. 2000). Our previous research had shown that the addition of *A. platensis* to the mice' diet 30 days prior to superovulation increases an activity of mitochondrial enzyme cytochrome C oxidase in the ovaries of treated animals (Kistanova et al. 2009b). In the present study maybe the cocktail of antioxidants present in *A. platensis* was the factor leading to more active and healthy oocyte maturation in the experimental groups of pigs. In support of the beneficial role of *A. platensis* for the developmental competence and oocyte quality are the results of Geffroy and Simon (2013). They reported that egg production in *A. platensis* diet-fed zebra fish was lower, but the survival rate of eggs was significantly higher (73% against 55% in control). Additionally, larvae from the experimental group developed

Fig. 2 Overview of a healthy corpus luteum (**a**, **c**) and a corpus albicans (**b**, **d**) in the ovary of a 180-day-old gilt treated for 120 days with 3 g SP per day. Bars represent 200 μm (**a**, **b**) and 100 μm (**c**, **d**)

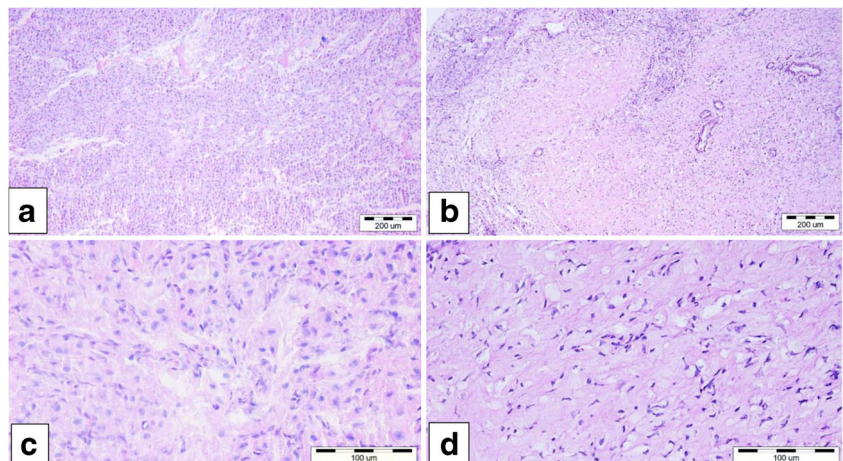


Table 4 Plasma levels of progesterone (nmol L⁻¹), estradiol (pmol L⁻¹), and ghrelin (pg mL⁻¹) at the time of slaughter

Hormones	Control	SP 2 g	SP 3 g
Progesterone	0.7 ± 0.2	2.1 ± 0.6 ^a	2.4 ± 0.7 ^b
Estradiol, E	283.8 ± 36.2	169.3 ± 45.9	205.4 ± 40.3
Ghrelin, G	94.0 ± 9.9	125.9 ± 9.3 ^c	115.6 ± 13.9

SP Spirulina platensis dosages

Gilts received either no treatment or daily treatment with 2 g or 3 g SP from the age of 60 days until slaughter at the age of 180 days. Data are expressed as mean ± SEM. The significance of differences between the groups was examined with Mann-Whitney *U* test (number *N* = 7 is too small to form a normal distribution)

^a *P* < 0.05 compared with the control group; *U* = 3, critical *U* for *P* < 0.05 is 8

^b *P* < 0.05 compared with the control group; *U* = 3, critical *U* for *P* < 0.05 is 8

^c *P* < 0.05 compared with the control group; *U* = 4, critical *U* for *P* < 0.05 is 8

faster, as a consequence of a significantly shorter hatching time. The same positive effect of *A. platensis* on hatching was observed in pheasants (Nikodémusz et al. 2010).

The novelty of obtained results is the establishment of the correlation between ghrelin and estradiol in the experimental groups of pigs, which demonstrated earlier and successful follicle development in the sexual maturation period. Many studies during the last decade suggest that ghrelin participates in the control of key aspects of reproductive function, including puberty onset (Fernandez-Fernandez et al. 2006; Muccioli et al. 2011; Tena-Sempere et al. 2013). Receptors of ghrelin and ghrelin are present in many peripheral tissues as well as in the ovary tissue (Gaytan et al. 2003). The previous studies with experimental animals have shown that estrogen is involved in the regulation of ghrelin secretion. So, the administration with estrogen decreased the number of ghrelin-producing cells and ghrelin mRNA levels in the stomach as well as the plasma ghrelin levels in ovariectomized rats (Matsubara et al. 2004). Related to the pigs, there are in vitro investigations showed that ghrelin increases estradiol secretion by modifying aromatase activity and it acts as an anti-apoptotic factor in pre-pubertal pig ovaries (Rak and Gregoraszczyk 2008). The studies in humans established the interaction between plasma levels of ghrelin and sex hormones (Rigamonti et al. 2002). Our findings also confirmed the presence of the relationship between ghrelin and progesterone in all groups. The correlation between ghrelin and estradiol was established in the experimental groups only. This allows supposing that, eventually, the SP additive affects the ovaries through ghrelin-estradiol pathways (Rak and Gregoraszczyk 2008).

In conclusion, the beneficial effect of *A. platensis* whole dry biomass resulted in successful follicular development and

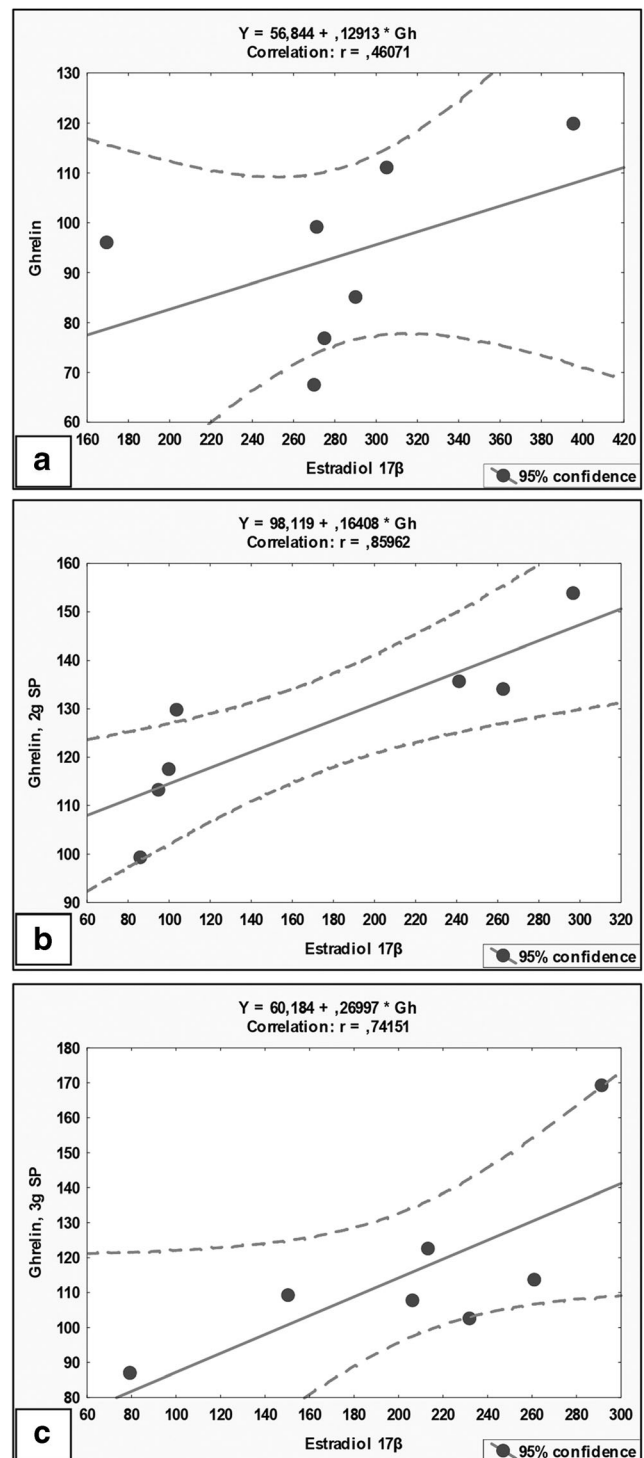


Fig. 3 Linear Pearson correlation analysis between ghrelin and 17-β-estradiol levels in 180-day-old gilts. **a** Control group. **b** Gilts treated daily for 120 days with 2 g SP. **c** Gilts treated daily for 120 days with 3 g SP

earlier ovulation in the ovary of experimental pigs during the sexual maturation period. The obtained results provided an evidence of the importance of metabolic hormone ghrelin for the establishment of a sex steroid profile during the puberty onset in the supplementation with *A. platensis* animals.

However, the exact mechanism directly or indirectly affecting *A. platensis* on folliculogenesis is not clear and future research in this direction still needs to be developed.

Acknowledgements We thank the COST Action FA1403 “POSITIVE” and the project DKOST 01/10 from the National Scientific Fund under the Bulgarian Ministry of Education and Science for the scientific and financial support of this research. The gift of *Spirulina platensis* from “Bioagrovet” (Lithuania) for this study is gratefully acknowledged.

Authors’ contribution EK coordinated the study and prepared the manuscript draft. KT was involved in the analysis of the data and the writing of the manuscript. NR, MY, and JG were responsible for the animal experiments, DA was responsible for the embedding and histological analysis of the ovarian tissues; CM and DJ performed the hormone analysis; And ASH and ASH developed the technology of growth of *Arthrospira* and performed dry biomass analysis and dose recommendation. All authors read and approved the final version of the manuscript.

Compliance with ethical standards All animals were reared and handled in accordance with the Bulgarian veterinary law related to the life conditions and welfare of livestock experimental animals (Regulation N20/01.11.2012 based on the Directive 2010/63/EU on the protection of animals used for scientific purposes in the Member States). The use of animals in the experiment was approved by the Ethics Board in Agricultural Institute-Shumen.

Conflict of interest The authors declare that they have no conflicts of interest.

References

- Ahmed SM, Hossain E, Kim GM, Hwang JA, Ji H, Yang CJ (2013) Effects of resveratrol and essential oils on growth performance, immunity, digestibility and fecal microbial shedding in challenged piglets. *Asian Australas J Anim Sci* 26:683–690
- Abdulrahman N, Hamad A (2014) Replacement of fishmeal with microalgae *Spirulina* on common carp weight gain, meat and sensitive composition and survival. *Pakistan J Nutr* 13:93–98
- Agarwal A, Gupta S, Sekhon L, Shah R (2008) Redox considerations in female reproductive function and assisted reproduction: from molecular mechanisms to health implications. *Antioxid Redox Signal* 10:1375–1403
- Belay A (2002) The potential application of *Spirulina* (*Arthrospira*) as a nutritional and therapeutic supplement in health management. *J Am Nutraceut Assoc* 5:26–49
- Belay A, Kato T, Ota Y (1996) *Spirulina* (*Arthrospira*): potential application as an animal feed supplement. *J Appl Phycol* 8:303–311
- Costa LB, Tse MLP, Miyada VS (2007) Herbal extracts as alternatives to antimicrobial growth promoters for newly weaned piglets. *Rev Bras de Zootec* 36:589–595
- Dumollard R, Ward Z, Carroll J, Duchon MR (2007) Regulation of redox metabolism in the mouse oocyte and embryo. *Development* 134:455–465
- Fernandez-Fernandez R, Martini AC, Navarro VM, Castellano JM, Dieguez C, Aguilar E, Pinilla L, Tena-Sempere M (2006) Novel signals for the integration of energy balance and reproduction. *Mol Cell Endocrinol* 254–255:127–132
- Gaytan F, Barreiro ML, Chopin LK, Herington AC, Morales C, Pinilla L, Casanueva FF, Aguilar E, Diéguez C, Tena-Sempere M (2003) Immunolocalization of ghrelin and its functional receptor, the type 1a growth hormone secretagogue receptor, in the cyclic human ovary. *J Clin Endocrinol Metab* 88:879–887
- Geffroy B, Simon O (2013) Effects of a *Spirulina platensis*-based diet on zebrafish female reproductive performance and larval survival rate. *Cybio* 37:31–38
- Muccioli G, Lorenzini T, Lorenzini M, Gheba C, Amoletti E, Rasoch GM, Castellucci M, Gualillo O, Melic R (2011) Beyond the metabolic role of ghrelin: a new player in the regulation of reproductive function. *Peptides* 32:2514–2521
- Govoni N, De Iasio R, Cocco C, Parmeggiani A, Galeati G, Bagottoc U, Brancia C, Spinaci M, Tamanini C, Pasquali R, Ferri GL, Seren E (2005) Gastric immunolocalization and plasma profiles of acyl-ghrelin in fasted and fasted-refed prepubertal gilts. *J Endocrinol* 186:505–513
- Holman BWB, Malau-Aduli AEO (2013) *Spirulina* as a livestock supplement and animal feed. *J Animal Physiol Animal Nutr* 97:615–623
- Hugh WI, Dominy W, Duerr E (1985) Evaluation of dehydrated spirulina (*Spirulina platensis*) as a protein replacement in swine starter diets. *Coll Trop Agric Human Res Univ Hawaii Res Ext Ser* 056:10
- Kistanova E, Kacheva D, Shumkov K, Abadjieva D, Borjaev G, Nevitov M, Shimkus A (2009a) Effect of biological active substances on the mouse embryo production *in vivo*. *Compt Rend l’Acad Bulgare Sci* 62:499–506
- Kistanova E, Marchev Y, Nedeva R, Kacheva D, Shumkov K, Georgiev B, Shimkus A (2009b) Effect of the *Spirulina platensis* included in the main diet on the boar sperm quality. *Biotechnol Animal Husbandry* 25:547–557
- Li P, Piao X, Ru Y, Han X, Xue L, Zhang H (2012) Effects of adding essential oil to the diet of weaned pigs on performance, nutrient utilization, immune response and intestinal health. *Asian-Aust. J Anim Sci* 25:1617–1626
- Mariey Y, Samak AHR, Ibrahim MA (2012) Effect of using *Spirulina platensis* algae as a feed additive for poultry diets: 1-productive and reproductive performances of local laying hens. *Egypt. Poult Sci* 32:201–215
- Matsubara M, Sakata I, Wada R, Yamazaki M, Inoue K, Sakai T (2004) Estrogen modulates ghrelin expression in the female rat stomach. *Peptides* 25:289–297
- Namkung H, Li M, Gong J, Yu H, Cottrill M, de Lange CFM (2004) Impact of feeding blends of organic acids and herbal extracts on growth performance, gut microbiota and digestive function in newly weaned pigs. *Can J Anim Sci* 84:697–704
- Nedeva R, Jordanova G, Kistanova E, Shumkov K, Georgiev B, Abadjieva D, Kacheva D, Shimkus A, Shimkine A (2014) Effect of the addition of *Spirulina platensis* on the productivity and some blood parameters on growing pigs. *Bulg. J Agric Sci* 20:680–684
- Nikodémusz E, Páskai P, Tóth L, Kozák J (2010) Effect of dietary *Spirulina* supplementation on the reproductive performance of farmed pheasants. *Technical Articles-Poultry Industry*, pp. 1–2
- Oetting LL, Utiyama CE, Giani PA, Ruiz UD, Miyada VS (2006) Effects of herbal extracts and antimicrobials on apparent digestibility, performance, organs morphometry and intestinal histology of weanling pigs. *Braz J Anim Sci* 35:1389–1397
- Peiretti PG, Meineri G (2011) Effects of diets with increasing levels of *Spirulina platensis* on the carcass characteristics, meat quality and fatty acid composition of growing rabbits. *Livest Sci* 140:218–224
- Platel K, Srinivasan K (2004) Digestive stimulant action of spices: a myth or reality? *Indian J Med Res* 119:167–179
- Rak A, Gregoraszczuk E (2008) Local feedback loop of ghrelin-GH in the pig ovary: action on estradiol secretion, aromatase activity and cell apoptosis. *Grow Horm. IGF Res* 18:221–227
- Rak-Mardyla A, Gregoraszczuk EL, Karpeta A, Duda M (2012) Expression of ghrelin and the ghrelin receptor in different stages of porcine corpus luteum development and the inhibitory effects of

- ghrelin on progesterone secretion, 3- β -hydroxysteroid dehydrogenase activity and protein expression. *Theriogenology* 77:1505–1512
- Rigamonti AE, Pincelli AI, Corra B, Viarengo R, Bonomo SM, Galimberti D, Scacchi M, Scarpini E, Cavagnini F, Muller EE (2002) Plasma ghrelin concentrations in elderly subjects: comparison with anorexic and obese patients. *J Endocrinol* 175:R1–R5
- Rudic V, Bulimaga V, Chiriac T, Cepoi L, Rudi L, Granaci V, Darie G, Toderash I, Macari V (2008) BioR – a new preparation from *Spirulina* biomass for reproductive function regulation of sire bulls and boars. *Proceed Intern Conf, Bucharest, 1–3 July, 2008*, <http://www.bioing.com>
- Saeid A, Chojnacka K, Korczyński M, Korniewicz D, Dobrzański Z (2013) Biomass of *Spirulina maxima* enriched by biosorption process as a new feed supplement for swine. *J Appl Phycol* 25:667–675
- Sánchez M, Bernal-Castillo J, Rozo C, Rodríguez I (2003) *Spirulina* (*Arthrospira*): an edible microorganism: a review. *Universitas Scientiarum* 8(1): 7–24.
- Sandev S (1979) Chemical methods for analysis of feed. Zemizdat, Sofia, p 182
- Shkolnik K, Tadmor A, Ben-Dor S, Nevo N, Galiani D, Dekel N (2011) Reactive oxygen species are indispensable in ovulation. *Proc Natl Acad Sci U S A* 108:1462–1467
- Simkus A, Martinavicius V, Kulpus J, Simkiene A, Knietkute N, Stankeviciene M (2008) The effect of microalgae *Spirulina platensis* on physiological processes and productivity of fattening pigs. *J Anim Sci* 2:36–40
- Shimkus A, Shimkiene A, Juozaitiene V, Zavodnik L, Juozaitis A, Muzikevicius A (2009) Influence of blue algae *Spirulina platensis* on the productivity of sows. *Compt Rend l'Acad Bulgare Sci* 62: 405–410
- Šimkus A, Šimkienė A, Černauskienė J, Kvietkutė N, Černauskas A, Paleckaitis M, Kerzienė S (2013) The effect of blue algae *spirulina platensis* on pig growth performance and carcass and meat quality. *Veterinarija ir zootechnika (Vet Med Zoot)* 61:70–74
- Sirotkin AV, Meszarosova M, Grossmann R, Benco A, Valenzuela F (2011) Effect of inhibitor and activator of ghrelin receptor (GHS-R1a) on porcine ovarian granulosa cell functions. *Gen Comp Endocrinol* 173:105–110
- Tena-Sempere M (2007) Roles of ghrelin and leptin in the control of reproductive function. *Neuroendocrinology* 86:229–241
- Tena-Sempere M, Benso A, Casanueva FF, Ghigo E, Granata A (2013) Ghrelin, the gonadal axis and the onset of puberty. In: Benso A, Casanueva FF, Ghigo E, Granata R (eds) *The ghrelin system* Karger, Basel 25, pp 69–82. <https://doi.org/10.1159/000346055>
- Thacker P (2013) Alternatives to antibiotics as growth promoters for use in swine production: a review. *J Anim Sci Biotechnol* 4(1):35
- Torres-Rovira L, Pallares P, Vigo E, Gonzalez-Anover P, Sanchez-Sanchez R, Mallo F, Gonzalez-Bulnes A (2011) Plasma leptin, ghrelin and indexes of glucose and lipid metabolism in relation to the appearance of post-weaning oestrus in Mediterranean obese sows (Iberian pig). *Reprod Dom Anim* 46:558–560
- Whaley SL, Hedgpeth VS, Farin CE, Martus NS, Jayes FCL, Britt JH (2000) Influence of vitamin A injection before mating on oocyte development, follicular hormones and ovulation in gilts fed high-energy diets. *J Anim Sci* 78:1598–1607
- Windisch W, Schedle K, Plitzner C, Kroismayr A (2008) Use of phytogetic products as feed additives for swine and poultry. *J Anim Sci* 86:E140–E148
- Yan L, Meng QW, Kim IH (2012) Effect of an herb extract mixture on growth performance, nutrient digestibility, blood characteristics, and fecal microbial shedding in weanling pigs. *Livest Sci* 145:189–195
- Yuan Y, Wheeler MB, Krisner RL (2012) Disrupted redox homeostasis and aberrant redox gene expression in porcine oocytes contribute to decreased developmental competence. *Biol Reprod* 87(4):78
- Zhang W, Lei Z, Su J, Chen S (2008) Expression of ghrelin in the porcine hypothalamo–pituitary–ovary axis during the estrous cycle. *Anim Reprod Sci* 109:356–367